

A Dimension-Five Twisted-Convolution Identity from a Shintani Ray Packet

Draft for review

27 July 2026

Abstract

For $K = \mathbb{Q}(\sqrt{3})$ and modulus $(5)\infty_2$, the ray group relevant to the dimension-five principal ghost is cyclic of order eight. We compute all twenty-four nonzero characteristics, their positive Kopp lifts, ray classes, cocycle signs, and modular multipliers. We then prove unconditionally that the four positive double-sine values form the reciprocal degree-eight ray-unit packet previously recognized numerically by Kopp. The proof specializes Shintani's 1978 algebraicity theorem, extracts the safe exponent 5760, certifies the labeled ray field and Frobenius orbit exactly, and uses rigorous Arb enclosures together with Voutier's height bound to remove the power. An exact calculation over $\mathbb{Q}(\zeta_5, \sqrt{6})$ then proves that all one hundred 2-minors of the reconstructed ghost matrix vanish. This establishes both predicted shifts for every dimension-five admissible tuple, and hence the complete dimension-five Twisted Convolution Conjecture. No Stark conjecture or **bnrstark** recognition is used as an input.

1 Canonical data and conventions

Put

$$K = \mathbb{Q}(\sqrt{3}), \quad \beta = 2 + \sqrt{3}, \quad Q = \langle 1, -4, 1 \rangle, \quad L = \begin{pmatrix} 4 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$B = L^3 = \begin{pmatrix} 56 & -15 \\ 15 & -4 \end{pmatrix}, \quad B \begin{pmatrix} 0 \\ 1/5 \end{pmatrix} - \begin{pmatrix} 0 \\ 1/5 \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \end{pmatrix}.$$

Label the real embeddings by $\infty_1(\sqrt{3}) = +\sqrt{3}$ and $\infty_2(\sqrt{3}) = -\sqrt{3}$. The modulus throughout is

$$\mathfrak{m} = (5)\infty_2.$$

To remove the standard double-sine ambiguity, define

$$S_2(z \mid \omega_1, \omega_2) := \frac{\Gamma_2(z \mid \omega_1, \omega_2)}{\Gamma_2(\omega_1 + \omega_2 - z \mid \omega_1, \omega_2)}. \quad (1)$$

Thus S_2 is the reciprocal of Kopp's $\text{Sin}_{2, \text{Kopp}}$ convention. For $0 \leq c \leq 5$ and $0 \leq b < 5$, write

$$\mathcal{D}_{c,b} := S_2\left(\frac{c + b\beta}{5} \mid \beta, 1\right).$$

For $(p, q) \neq (0, 0)$, let $r \equiv -p - q \pmod{5}$. The principal-form specialization of the AFK three-factor formula is

$$\tilde{\nu}_{p,q} = (-1)^{5(p+q)+pq+\min(5,p+q)} \mathcal{D}_{5-p,q} \mathcal{D}_{5-r,p} \mathcal{D}_{5-q,r}. \quad (2)$$

Formula (2) is used only for nonzero characteristics. Later we set the table entry $T_{0,0} := \sqrt{6}$ solely so that the identity Weyl coefficient is $T_{0,0}/\sqrt{6} = 1$. This reconstruction convention is not AFK's auxiliary zero-characteristic cocycle value.

Lemma 1 (Derivation and convention check for (2)). *Formula (2) is the specialization of AFK Definition 1.32 to $t = (5, 1, \langle 1, -4, 1 \rangle)$, with no unrecorded reciprocal or sign.*

Proof. AFK Definition 1.32 first forms $\tilde{\nu}_p = \Phi_p \text{shin}_B^{p/5}(\beta)$. Insert the defining modular-to-Jacobi quotient [2, equation (1.26)], and then use the Jacobi cocycle decomposition [2, equations (8.2)–(8.5)] and the S -kernel formula [2, equation (8.6)]. In the notation of AFK Section 8.2,

$$L = T^4 S, \quad B = L^3, \quad L\beta = \beta, \quad j_{L^k}(\beta) = \beta^k.$$

Thus the word $B = (T^4 S)^3$ contributes exactly three S -kernels. Modulo five the characteristic orbit is

$$(p, q) \mapsto (r, p) \mapsto (q, r) \mapsto (p, q), \quad r \equiv -p - q \pmod{5},$$

and the three unreduced linear arguments are

$$\frac{q\beta - p}{5}, \quad \frac{p\beta - r}{5}, \quad \frac{r\beta - q}{5}.$$

Reducing their first coordinates to the representatives $5 - p, 5 - r, 5 - q$ gives respectively $\mathcal{D}_{5-p,q}$, $\mathcal{D}_{5-r,p}$, and $\mathcal{D}_{5-q,r}$. AFK's S -kernel is its displayed double sine in the denominator (AFK Section 8.2, equation for σ_S). That double sine is Kopp's $\text{Sin}_{2,\text{Kopp}}$; hence each denominator is precisely one factor of the reciprocal convention (1).

For completeness, apply the integral-translation law [2, equation (8.9)] to the three representative changes and multiply by the phase in AFK Definition 1.30. Direct substitution gives

$$\Phi_{p,q}(\text{three translation multipliers}) = (-1)^{E(p,q)}, \quad E(p,q) = 5(p+q) + pq + \min(5, p+q).$$

This is a finite parity calculation:

$$E(p,q) \equiv \begin{cases} pq & (p+q \leq 5), \\ p+q+pq+1 & (p+q \geq 5) \end{cases} \pmod{2},$$

where the two expressions agree at $p+q = 5$. In the order $p = 0, \dots, 4$ (columns $q = 0, \dots, 4$), the resulting signs are

$$\begin{pmatrix} * & + & + & + & + \\ + & - & + & - & + \\ + & + & + & + & - \\ + & - & + & + & + \\ + & + & - & + & - \end{pmatrix};$$

the asterisk is the excluded zero characteristic. This accounts for all twenty-four cases and proves (2) directly from the cited definitions. $\square$

2 Ray labels and all characteristics

Lemma 2 (Certified ray arithmetic). *The class and ray groups are*

$$\text{Cl}(K) = 1, \quad \text{Cl}_{(5)}(K) \cong C_4, \quad \text{Cl}_{\mathfrak{m}}(K) \cong C_8, \quad \text{Cl}_{(5)\infty_1\infty_2}(K) \cong C_8 \times C_2. \quad (3)$$

Kopp's residue/sign class $\mathfrak{R}$ is the element $4 \in C_8$. The fiber of $\text{Cl}_{(5)\infty_1\infty_2}(K) \rightarrow \text{Cl}_{\mathfrak{m}}(K)$ has order two, so the exponent in Kopp's theorem is $n = 2/2 = 1$. The computation is unconditional.

Proof. PARI/GP 2.15.4 gives the structures in (3), and `bnfcertify=1`. The unit group is $\{\pm\beta^n : n \in \mathbb{Z}\}$ and $\beta^3 \equiv 1 \pmod{5}$. Hence units congruent to -1 modulo 5 are $-\beta^{3k}$. Their value under ∞_2 is $-(2 - \sqrt{3})^{3k} < 0$, so no global unit realizes the pair $(-1 \pmod{5}, + \text{ at } \infty_2)$ defining Kopp's $\mathfrak{R}$. It is therefore the nontrivial sign class, namely the unique element of order two in C_8 . $\square$

Proposition 3 (No quadratic-character shortcut). *The Kopp partial-zeta difference has zero coefficient at the unique order-two character of $\text{Cl}_m(K)$. It is supported on four characters, all of order eight.*

Proof. Write $\text{Cl}_m(K) = \langle g \rangle \cong C_8$ and $\mathfrak{R} = g^4$. For $\chi_k(g) = \exp(2\pi i k/8)$, Fourier inversion gives

$$\zeta_m(s, \mathfrak{A}) - \zeta_m(s, \mathfrak{R}\mathfrak{A}) = \frac{1}{8} \sum_{k=0}^7 \overline{\chi_k(\mathfrak{A})} (1 - \overline{\chi_k(\mathfrak{R})}) L(s, \chi_k). \quad (\text{CF})$$

Since $\chi_k(\mathfrak{R}) = (-1)^k$, the coefficient is zero for even k and two for odd k . The support is therefore $k = 1, 3, 5, 7$, whose characters all have order eight. The unique quadratic character is χ_4 ; it is even on $\mathfrak{R}$ and its coefficient in (CF) is zero. Hence the difference cannot factor through the quadratic quotient. This is the precise reason the dimension-four quadratic L -function reduction does not recur. Equivalently, $\mathfrak{R} = g^4$ lies in the kernel $\langle g^2 \rangle$ of the quotient $C_8 \rightarrow C_2$, so $\mathfrak{A}$ and $\mathfrak{R}\mathfrak{A}$ have the same image in the unique quadratic quotient and their difference projects to zero. $\square$

For each $(p, q) \neq (0, 0)$ start with p and repeatedly subtract five; let $\tilde{p}$ be the first lift for which

$$q(2 - \sqrt{3}) - \tilde{p} > 0. \quad (4)$$

Let $\mathfrak{q}$ be the prime above 3, and use it as the generator of C_8 . PARI's raw cyclic basis g gives the discrete logarithms of the positive principal ideals $(q\beta - \tilde{p})$ in the audit transcript, with $\mathfrak{q} = g^5$. Multiplication by 5, which is its own inverse modulo 8, converts those raw logs to the normalized $\mathfrak{q}$ -coordinates below:

$p \backslash q$	0	1	2	3	4
0	–	0	6	2	4
1	4	7	5	7	0
2	2	5	5	6	3
3	6	7	2	1	1
4	0	4	3	1	3

(5)

The corresponding positive lifts are 0 in the first row, -4 in row one except at $(1, 4)$ where it is 1, and $-3, -2, -1$ in rows two, three, and four.

Lemma 4 (All multiplier comparisons). *For every nonzero $(p, q) \bmod 5$, Kopp's modular multiplier equals the square of the AFK phase at that characteristic.*

Proof. Write $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{r} = (\tilde{p}/5, q/5)^\top$, with $\tilde{p}$ chosen by (4). Substitution in Kopp's defining theta-character gives the exponent

$$\frac{1}{2} \{ (c - d + 1)r_1 + (-a + b + 1)r_2 - cdr_1^2 + 2(a - 1)dr_1r_2 - (a - 2)br_2^2 \}. \quad (6)$$

For $B = \begin{pmatrix} 56 & -15 \\ 15 & -4 \end{pmatrix}$, direct reduction of (6) modulo one gives

$$\frac{p^2 - 4pq + q^2}{5} \pmod{1} \quad (7)$$

in all twenty-four cases. Also $s(-4, 15) = -19/90$, so the Rademacher invariant is three and $\psi^2(B) = i$. Therefore the Kopp multiplier has exponent

$$-\frac{1}{4} - \frac{p^2 - 4pq + q^2}{5} \pmod{1}, \quad (8)$$

which is precisely the exponent of the square of the AFK phase. The exact twenty-four-row substitution, including every positive lift, is part of the bridge certificate. $\square$

For a normalized ray log k , name the positive square root of its ray value by

k	0	1	2	3	4	5	6	7
positive root	x	w	y^{-1}	z	x^{-1}	w^{-1}	y	z^{-1} .

The class log in (5) selects the unsigned entry and the explicit sign in (2) selects its sign. Thus the complete reconstruction table is

$$T_5 = \begin{pmatrix} \sqrt{6} & x & y & y^{-1} & x^{-1} \\ x^{-1} & -z^{-1} & w^{-1} & -z^{-1} & x \\ y^{-1} & w^{-1} & w^{-1} & y & -z \\ y & -z^{-1} & y^{-1} & w & w \\ x & x^{-1} & -z & w & -z \end{pmatrix}. \quad (9)$$

Here x, y, z, w denote the four positive representatives of the ray-class pairs

$$(0, 4), \quad (6, 2), \quad (3, 7), \quad (1, 5), \quad (10)$$

respectively. In particular, the table is generated from the ray logs and the sign exponent; it is not stored as a list of expected answers in the bridge generator.

3 The analytic ray packet

Set

$$\begin{aligned} P(U) = & U^8 - (8 + 5\sqrt{3})U^7 + (53 + 30\sqrt{3})U^6 \\ & - (156 + 90\sqrt{3})U^5 + (225 + 130\sqrt{3})U^4 \\ & - (156 + 90\sqrt{3})U^3 + (53 + 30\sqrt{3})U^2 \\ & - (8 + 5\sqrt{3})U + 1. \end{aligned} \quad (11)$$

It is the reciprocal lift $P(U) = U^4 F(U + U^{-1})$ of

$$\begin{aligned} F(V) = & V^4 - (8 + 5\sqrt{3})V^3 + (49 + 30\sqrt{3})V^2 \\ & - (132 + 75\sqrt{3})V + (121 + 70\sqrt{3}). \end{aligned} \quad (12)$$

Kopp previously recognized the same polynomial numerically as the Stark-unit polynomial for $\mathbb{Q}(\sqrt{3})$ and modulus five; see [4, §7, (7.21)] and [5, Example 1.17, (3.12)]. PARI's `bnrstark` reproduces (12), but that recognition is not used as a proof below.

The square roots of the reciprocal packet have absolute polynomial

$$\begin{aligned} Q(X) = & X^{32} - 16X^{30} + 95X^{28} - 260X^{26} + 355X^{24} - 348X^{22} \\ & + 388X^{20} - 300X^{18} + 195X^{16} - 300X^{14} + 388X^{12} \\ & - 348X^{10} + 355X^8 - 260X^6 + 95X^4 - 16X^2 + 1. \end{aligned} \quad (13)$$

Define $w^{-1}, z^{-1}, x^{-1}, y^{-1}, y, x, z, w$ to be the unique positive roots of (13) in, respectively,

$$\begin{aligned} & (0.424835, 0.424836), (0.481476, 0.481477), (0.506963, 0.506964), (0.782390, 0.782391), \\ & (1.278133, 1.278134), (1.972526, 1.972527), (2.076946, 2.076947), (2.353849, 2.353850). \end{aligned} \quad (14)$$

Sturm's theorem certifies exactly one root in every interval.

3.1 Shintani's theorem with an explicit exponent

For a ray class $A \in \text{Cl}_{\mathfrak{m}}(K)$ put

$$X_A := \exp Z'_{\mathfrak{m}}(0, A).$$

Lemma 5 (Specialized Shintani algebraicity). *For every $A \in \text{Cl}_{\mathfrak{m}}(K)$,*

$$X_A^{5760} \in \mathcal{O}_H^\times,$$

where H is the ray field of modulus $\mathfrak{m}$. These powers have the Artin action indexed by the ray classes. At every embedding of H inducing ∞_2 on K , their absolute value is one.

Proof. Work first in the full narrow ray group

$$C := \text{Cl}_{(5)\infty_1\infty_2}(K) \cong C_8 \times C_2.$$

In Shintani's notation, let μ be the class represented by $11 - 5\beta$, which is congruent to 1 modulo 5 and has signs $(-, +)$ at (∞_1, ∞_2) . Let ν be the class represented by 4, which is congruent to -1 modulo 5 and is totally positive. The exact ray logs are

$$\mu = [4, 1] \in C, \quad \nu = [4, 0] \in C.$$

Forgetting ∞_1 kills μ and sends ν to the unique order-two element $4 \in C_8$. Hence

$$G = \langle \mu \rangle, \quad \nu \notin G, \quad C/G \simeq \text{Cl}_{\mathfrak{m}}(K).$$

Every unit is $\pm\beta^n$, and β has order three modulo 5. Thus no totally positive unit is -1 modulo 5, which is Shintani's condition (0-3). There is no norm- -1 unit, since $a^2 - 3b^2 = -1$ is impossible modulo 3; this implies condition (0-6).

The field H has signature $(8, 4)$ and a unique octic subfield

$$M \simeq \mathbb{Q}(\zeta_{60})^+.$$

Thus H/M is quadratic, M is the maximal absolutely abelian subfield of H , and exactly one real place of K splits in H . These are Shintani's hypothesis (0-9). His Theorem 2 [6] therefore applies to $X_{(5)}(c, G)$.

This invariant is exactly the one-place zeta difference used here. Indeed, if c maps to A , then the inverse images of A and RA in C are respectively

$$\{c, c\mu\}, \quad \{c\nu, c\mu\nu\}.$$

Shintani's Theorem 1 consequently gives

$$\begin{aligned} \log X_{(5)}(c, G) &= \zeta'_{\text{full}}(0, c) - \zeta'_{\text{full}}(0, c\nu) \\ &\quad + \zeta'_{\text{full}}(0, c\mu) - \zeta'_{\text{full}}(0, c\mu\nu) \\ &= \zeta'_{\mathfrak{m}}(0, A) - \zeta'_{\mathfrak{m}}(0, RA) = Z'_{\mathfrak{m}}(0, A). \end{aligned}$$

It remains to make Shintani's existential power explicit. The normal closure of H is the full two-infinite-place ray field. It is also the degree-sixteen ray field over $\mathbb{Q}(\sqrt{-5})$ with conductor

$$\mathfrak{c} = \begin{pmatrix} 15 & 0 \\ 0 & 3 \end{pmatrix}$$

in PARI's integral basis. The quadratic-subfield enumeration contains a unique copy of $\mathbb{Q}(\sqrt{-5})$, so this identification preserves the imaginary quadratic field occurring in Shintani's proof.

Shintani's Proposition 4 expresses the real invariant as a product of ratios of imaginary-quadratic W -invariants with integral multiplicities. For a nontrivial divisor $\mathfrak{d} \mid \mathfrak{c}$, clearing the absolute-value square root and the distribution exponent requires a multiple of $12f_{\mathfrak{d}}n(S)$. For the trivial divisor the required exponent is $12h_k n(S)$. The exact exponents for the eight divisors are

$$384, 288, 288, 144, 240, 360, 360, 180,$$

whose least common multiple is 5760. Since (5) is prime and its only proper divisor fails (0-3), Shintani's $Y_{(5)}$ equals $X_{(5)}$ and no induction exponent is added. Proposition 5 supplies the unit, Artin, and nonsplit-modulus-one assertions. $\square$

3.2 The labeled candidate field

The absolute polynomial of a root of (11) is

$$\begin{aligned} p(T) = & T^{16} - 16T^{15} + 95T^{14} - 260T^{13} + 355T^{12} - 348T^{11} \\ & + 388T^{10} - 300T^9 + 195T^8 - 300T^7 + 388T^6 - 348T^5 \\ & + 355T^4 - 260T^3 + 95T^2 - 16T + 1. \end{aligned}$$

Lemma 6 (Exact field and Frobenius labels). *The field defined by (11) is the labeled ray field H over K . Let U be the root of p in*

$$(3.890, 3.891).$$

Then U is a norm-one unit. Arithmetic Frobenius at the prime $\mathfrak{q}$ above 3 acts transitively on its eight real conjugates, which lie successively in

$$\begin{aligned} & (3.890, 3.891), (5.540, 5.541), (0.612, 0.613), (4.313, 4.314), \\ & (0.257, 0.258), (0.180, 0.181), (1.633, 1.634), (0.231, 0.232). \end{aligned}$$

At the embeddings above ∞_2 , all conjugates of U have absolute value one.

Proof. Applying `rnfequation` to both the candidate and the ray field, `nfisism` returns eight maps. Exact substitution shows that all eight maps fix the labeled root of $y^2 - 4y + 1$; hence the isomorphism is over K , not merely over $\mathbb{Q}$. Moreover `bnfcertify=1`, `bnfisunit` gives an exact unit decomposition, and the norm of U is one.

The prime $\mathfrak{q}$ generates the ray group. Modulo $\mathfrak{q}$ the relative polynomial is

$$T^8 + T^7 + 2T^6 + 2T^2 + T + 1 \in \mathbb{F}_3[T],$$

which is irreducible. Arithmetic Frobenius is therefore $T \mapsto T^3$. Reduction modulo this polynomial identifies one of the exact `nfgaloisconj` automorphisms with Frobenius.

A rational interval of width 10^{-80} contains exactly one root of p in $(3.890, 3.891)$. Exact rational interval evaluation of the successive Frobenius polynomials places their images inside the eight disjoint intervals in the statement. This proves the labels Galois-equivariantly.

Finally, under the nonidentity embedding of K , substitute $T = 2(Y^2 - 1)/(Y^2 + 1)$ in the trace polynomial (12). The exact `nfpolsturm` counts are $[8, 0]$, so all four trace roots at that embedding lie in $(-2, 2)$. Each corresponding reciprocal quadratic has both roots on the unit circle. $\square$

3.3 Certified removal of the power

Theorem 7 (Unconditional ray packet). *For*

$$\mathcal{X}_k := \exp Z'_m(0, \mathfrak{q}^k), \quad 0 \leq k < 8,$$

the exact Artin-labeled packet is

$$(\mathcal{X}_0, \dots, \mathcal{X}_7) = (x^2, w^2, y^{-2}, z^2, x^{-2}, w^{-2}, y^2, z^{-2}). \quad (15)$$

In particular, $\mathcal{X}_0 = U$.

Proof. Kopp's proved Kronecker-limit theorem, with the convention and multiplier checked above, gives

$$\tilde{\nu}_{p,q}^2 = \exp Z'_m(0, \mathfrak{A}_{p,q}); \quad (16)$$

here $\mathfrak{A}_{p,q}$ is the normalized ray class in (5). The four positive generators at characteristics

$$(0, 1), \quad (0, 2), \quad (2, 4), \quad (3, 3)$$

have $\mathfrak{q}$ -coordinates 0, 6, 3, 1, respectively.

We evaluate the regularized integral for the reciprocal double sine (1) with Arb balls through `python-flint` [9]. Every Simpson panel uses an Arb enclosure of the fourth derivative. All end-points, widths, tolerances, and acceptance comparisons are exact rationals or outward-rounded balls. Near zero, the positive series of $H_c(t) = \sinh(ct/2)/(ct/2)$ gives the rational bound

$$|f(t) - f(0)| \leq 4t^2 \quad (0 \leq t \leq 10^{-4}).$$

For the tail, every reduced argument and its complement belongs to $(1/100, 5)$; beyond 36, each exponential denominator exceeds $3/4$, giving the remainder bound $e^{-36}/9$. The candidate roots are selected by disjoint rational isolating intervals, independently of library ordering.

For each of the four characteristics, the ball for $\log(\tilde{\nu}_{p,q}^2) - \log(U_k)$ contains zero. The largest radius is less than

$$\epsilon = 2.23 \times 10^{-9}.$$

The reciprocal partners give the remaining four real embeddings.

By Lemma 5, both $\mathcal{X}_0^{5760}$ and U^{5760} are units in the same labeled field. Set

$$\eta = \mathcal{X}_0^{5760}/U^{5760}.$$

Lemma 6 and the interval labels show that at every real embedding,

$$|\log |\sigma(\eta)|| < 5760\epsilon < 1.29 \times 10^{-5}.$$

At the four complex pairs, both numerator and denominator have absolute value one. Hence

$$h(\eta) < 1.29 \times 10^{-5}.$$

If $3 \leq d = [\mathbb{Q}(\eta) : \mathbb{Q}] \leq 16$ and η is not a root of unity, Voutier's explicit bound [7] gives

$$h(\eta) > \frac{1}{4d} \left(\frac{\log \log d}{\log d} \right)^3.$$

Its minimum for $3 \leq d \leq 16$ is $5.2279533222 \dots \times 10^{-5}$, a contradiction. A non-torsion quadratic unit has height at least $\frac{1}{2} \log((1 + \sqrt{5})/2)$, and a rational unit is ± 1 . Thus η is a root of unity. Since H has a real embedding, its only roots of unity are ± 1 ; positivity at the distinguished embedding gives $\eta = 1$. Therefore $\mathcal{X}_0^{5760} = U^{5760}$ and positivity yields $\mathcal{X}_0 = U$. The exact Artin action in Lemma 5, together with the certified Frobenius orbit in Lemma 6, proves the full packet (15). $\square$

Since every reciprocal double-sine factor in (2) is positive at the displayed arguments, equations (2), (5), and (15) now derive the signed table (9) unconditionally.

4 Exact finite cancellation

Let $\zeta_5 = e^{2\pi i/5}$ and $\tau = -e^{\pi i/5} = \zeta_5^3$. Define, without reference to software,

$$K_{r,c} := \frac{1}{5\sqrt{6}} \sum_{\substack{0 \leq a,b < 5 \\ r \equiv c+a \pmod{5}}} T_{a,b} \tau^{ab} \zeta_5^{bc}, \quad 0 \leq r, c < 5, \quad (17)$$

where T is (9). Equivalently, $K = \frac{1}{5} \sum_{a,b} (T_{a,b}/\sqrt{6}) D_{a,b}$ for $D_{a,b} e_c = \tau^{ab} \zeta_5^{bc} e_{c+a}$. In particular,

$$\text{Tr } K = T_{0,0}/\sqrt{6} = 1. \quad (18)$$

Proposition 8 (Exact 100-minor certificate). *For the algebraic values selected by (13)–(14), all one hundred 2-minors of the matrix (17) vanish exactly. Hence $\text{rank } K = 1$ and $K^2 = K$.*

Proof. Over the coefficient field $\mathbb{Q}(\zeta_5, \sqrt{6})$, (13) has four degree-eight factors. Let w be the interval-defined root in (14). Exact rational interval arithmetic applied to the `nfgaloisconj` polynomials proves that `nfgaloisconj` substitutions are

	w	w^{-1}	y	y^{-1}	x	x^{-1}	z	z^{-1}
conjugate index	2	8	5	4	10	6	16	12.

(19)

with every image lying inside its corresponding interval in (14). Moreover, exact `nfisincl` identities express $\sqrt{5}$ and $\sqrt{6}$ as polynomials in w ; rational interval evaluation selects their positive signs. Among the four factors, factor four is the unique factor on which both of these subfield elements equal the positive $\sqrt{5}$ and positive $\sqrt{6}$. This certifies, rather than numerically assumes, that the interval-defined tuple is the tuple used in the factor-four calculation.

Substitution in (9) and (17), followed by exact reduction modulo the selected factor, makes all one hundred minors zero. As a factor selection check, factors one, two, and three leave respectively 70, 100, 100 nonzero minors. This is exact number-field arithmetic, not floating-point evaluation. The complete script and transcript are supplied with the manuscript.

The matrix is nonzero because (18) gives trace one. Thus its rank is exactly one. Any rank-one matrix satisfies $K^2 = (\text{Tr } K)K$, proving idempotency. $\square$

5 Dimension-five twisted convolution

Lemma 9 (Weyl coefficient bridge). *The matrix K of (17) is the standard-basis Weyl reconstruction of the canonical AFK ghost for $t = (5, 1, \langle 1, -4, 1 \rangle)$. Moreover, $K^2 = K$ is equivalent to the $\lambda = 1$ instance of AFK equation (1.49):*

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \zeta_5^{\langle \mathbf{p}, (I+L_{z,t})\mathbf{q} \rangle} \text{shin}_{A_t}^{\mathbf{q}/5}(\beta) \text{shin}_{A_t^{-1}}^{(\mathbf{q}-\mathbf{p})/5}(\beta) = 25\delta_{\mathbf{p}, \mathbf{0}}. \quad (20)$$

Proof. For $\mathbf{p} \neq \mathbf{0}$, put $\nu_{\mathbf{p}} := \tilde{\nu}_{\mathbf{p}}$ as given by (2). The first two unconditional parts of AFK's construction give

$$\nu_{\mathbf{p}} \nu_{-\mathbf{p}} = 1. \quad (21)$$

Theorem 7, followed by (2) and (5), identifies these twenty-four numbers with the nonzero entries of T . The actual ghost reconstruction is therefore

$$K = \frac{1}{5}I + \frac{1}{5\sqrt{6}} \sum_{\mathbf{p} \neq \mathbf{0}} \nu_{\mathbf{p}} D_{\mathbf{p}}. \quad (22)$$

The artificial entry $T_{0,0} = \sqrt{6}$ packages the first term of (22) into the single entry formula (17); it is not an analytic zero-characteristic value.

Now expand $K^2 - K$ using $D_{\mathbf{p}}D_{\mathbf{q}} = \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} D_{\mathbf{p}+\mathbf{q}}$ and compare every Weyl coefficient. For $\mathbf{p} \neq \mathbf{0}$, multiplication by 150 makes its coefficient

$$\sum_{\substack{\mathbf{q} \bmod 5 \\ \mathbf{q} \neq \mathbf{0}, \mathbf{p}}} \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} \nu_{\mathbf{p}-\mathbf{q}} \nu_{\mathbf{q}} - 3\sqrt{6} \nu_{\mathbf{p}}. \quad (23)$$

Using (21), the product in (23) is $\nu_{\mathbf{q}}/\nu_{\mathbf{q}-\mathbf{p}}$.

AFK's $\nu_{\mathbf{0}}^{\text{aux}}$ is an auxiliary cocycle value, not the identity coefficient in (22). Their zero-characteristic lemma specializes to

$$\nu_{\mathbf{0}}^{\text{aux}} + (\nu_{\mathbf{0}}^{\text{aux}})^{-1} = -(5-2)\sqrt{6} = -3\sqrt{6}. \quad (24)$$

After adjoining the two omitted endpoints $\mathbf{q} = \mathbf{0}, \mathbf{p}$, their combined contribution is

$$\nu_{\mathbf{p}}(\nu_{\mathbf{0}}^{\text{aux}} + (\nu_{\mathbf{0}}^{\text{aux}})^{-1}) = -3\sqrt{6} \nu_{\mathbf{p}}.$$

Thus (23) is exactly the full endpoint-corrected sum

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} \frac{\nu_{\mathbf{q}}}{\nu_{\mathbf{q}-\mathbf{p}}}, \quad (25)$$

with the two endpoint ratios interpreted using $\nu_{\mathbf{0}}^{\text{aux}}$.

Write $\Phi_{\mathbf{q}}$ for the AFK phase. Their phase-ratio calculation specializes, with $r = f_j/f = 1$ and $d_j = d = 5$, to

$$\frac{\Phi_{\mathbf{q}}}{\Phi_{\mathbf{q}-\mathbf{p}}} = (-1)^{1+s_5(\mathbf{p})} \tau^{Q(\mathbf{p})} \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} \zeta_5^{\langle \mathbf{p}, L_{z,t} \mathbf{q} \rangle}. \quad (26)$$

The first two factors are nonzero and independent of $\mathbf{q}$. Multiplying the remaining τ factor by the displacement chirp in (25), and using $\tau^2 = \zeta_5$, gives exactly

$$\zeta_5^{\langle \mathbf{p}, (I+L_{z,t}) \mathbf{q} \rangle}.$$

The cocycle inverse identity

$$\frac{\text{shin}_{A_t}^{\mathbf{q}/5}(\beta)}{\text{shin}_{A_t}^{(\mathbf{q}-\mathbf{p})/5}(\beta)} = \text{shin}_{A_t}^{\mathbf{q}/5}(\beta) \text{shin}_{A_t^{-1}}^{(\mathbf{q}-\mathbf{p})/5}(\beta) \quad (27)$$

gives the two factors printed in (20). Here $A_t = B$, $L_{z,t} = L$, and $f_t(1) = 2 + 9 \equiv 1 \pmod{5}$, so the identity twist produces exactly the chirp in (25). The remaining factor depends on $\mathbf{p}$ but is nonzero and independent of $\mathbf{q}$. Hence the vanishing of every nonzero Weyl coefficient is equivalent to (20) for $\mathbf{p} \neq \mathbf{0}$. For $\mathbf{p} = \mathbf{0}$, the cocycle inverse identity makes the sum 25 automatically. This is the specialization of AFK equations (5.78)–(5.98), including rather than suppressing their endpoint correction. $\square$

Theorem 10 (Complete dimension-five TCC). *For every dimension-five admissible tuple t ,*

$$0, 1 \in \mathcal{Z}_t.$$

Furthermore $\mathcal{Z}_{t'} = \mathcal{Z}_t$ for any two dimension-five admissible tuples t, t' .

Proof. Proposition 8 and Lemma 9 prove the $\lambda = 1$ identity for the canonical tuple. The twist condition is

$$\det(G)f_t(\lambda) \equiv 1 \pmod{5}, \quad f_t(\lambda) = 2\lambda + 9.$$

Thus $G = I$ gives $2\lambda + 9 \equiv 1 \pmod{5}$, or $\lambda \equiv 1 \pmod{5}$.

Complex conjugation preserves idempotency and sends λ to $\bar{\lambda} = -(\lambda + d - 1) = 1 - \lambda \pmod{5}$. It therefore sends the verified shift 1 to shift 0. Equivalently the conjugate twist has determinant -1 , and $-(2\lambda + 9) \equiv 1 \pmod{5}$ gives $\lambda \equiv 0 \pmod{5}$.

Finally, dimension-five admissibility forces $r = 1$, $K = \mathbb{Q}(\sqrt{3})$, and $j = m = 1$. Since the relevant conductor is one, every admissible form has discriminant twelve. The ordinary class number is one, so primitive indefinite forms of discriminant twelve constitute one $\mathrm{GL}_2(\mathbb{Z})$ -equivalence class. (The narrow class number is not needed: determinant- -1 transformations are allowed.) Thus every admissible form is integrally equivalent to the canonical one in the sense required here.

For clarity, the transport does not invoke AFK's conditional transformation theorem. If t_M is obtained from t by $M \in \mathrm{GL}_2(\mathbb{Z})$, then unconditionally

$$A_{t_M} = M^{-1}A_tM, \quad L_{z,t_M} = M^{-1}L_{z,t}M, \quad \beta_{t_M} = M^{-1}\beta_t.$$

Kopp's covariance formula sends each cocycle factor at $(t_M, \mathbf{q})$ to the corresponding factor at $(t, \ell M \mathbf{q})$, with $\ell = \mathrm{sgn}(j_{M^{-1}}(\beta_t))$; when $\det M = -1$ both factors are complex-conjugated, which does not change the real TCC sum. Reindexing

$$\mathbf{q}' = \ell M \mathbf{q}, \quad \mathcal{I}'_{\ell M \mathbf{p}} = \ell M \mathcal{I}_{\mathbf{p}},$$

and using symplectic covariance transforms the complete left side of (20) for $(t_M, \mathbf{p})$ into that for $(t, \ell M \mathbf{p})$. Hence $\mathcal{Z}_t \subseteq \mathcal{Z}_{t_M}$. Applying the same calculation to M^{-1} gives the reverse inclusion and therefore equality. $\square$

6 Scope and reproducibility

Theorem 10 proves the complete formal TCC in dimension five. It does not prove a dimension-five SIC, the TCC in arbitrary dimensions, or Zauner's conjecture. The new number-theoretic step is Theorem 7: it proves the previously conjectural dimension-five ray packet by combining Kopp's proved limit formula with the applicable case of Shintani's proved algebraicity theorem and an effective height argument.

The supplementary package contains:

- the 25-characteristic bridge certificate and generator;
- the exact C_8 Fourier-support audit excluding the quadratic character;
- the PARI transcript with `bnfcertify=1` and all positive ray-class logs;
- the Sturm root-isolation transcript;
- the exact interval/conjugate/factor embedding certificate;
- the formal 100-minor Laurent certificate; and
- the exact four-factor minor-reduction transcript;
- the Shintani sign-class, conductor, and exponent-5760 audit;
- the labeled K -isomorphism, unit, Sturm, and Frobenius-orbit certificate; and
- the rigorous Arb double-sine enclosure and height-gap transcript.

Their SHA-256 digests are recorded in `certificates/SHA256SUMS`.

References

- [1] G. Kopp, *The Shintani–Faddeev modular cocycle: Stark units from q -Pochhammer ratios*, arXiv:2411.06763.
- [2] D. M. Appleby, S. T. Flammia, and G. S. Kopp, *A constructive approach to Zauner’s conjecture via the Stark conjectures*, arXiv:2501.03970.
- [3] G. Kopp, *SIC-POVMs and the Stark conjectures*, Int. Math. Res. Not. IMRN 2021, no. 18, 13812–13838; arXiv:1807.05877.
- [4] G. S. Kopp, *Indefinite zeta functions*, Res. Math. Sci. **8**, 17 (2021), <https://doi.org/10.1007/s40687-021-00252-9>.
- [5] G. S. Kopp, *A Kronecker limit formula for indefinite zeta functions*, Res. Math. Sci. **10**, 24 (2023), <https://doi.org/10.1007/s40687-023-00384-0>.
- [6] T. Shintani, *On certain ray class invariants of real quadratic fields*, J. Math. Soc. Japan **30** (1978), 139–167, <https://doi.org/10.2969/jmsj/03010139>.
- [7] P. Voutier, *An effective lower bound for the height of algebraic numbers*, Acta Arith. **74** (1996), 81–95, <https://doi.org/10.4064/aa-74-1-81-95>.
- [8] The PARI Group, *PARI/GP version 2.15.4*, Université de Bordeaux, 2023, <https://pari.math.u-bordeaux.fr/>.
- [9] F. Johansson and contributors, *python-flint 0.9.0*, <https://github.com/flintlib/python-flint>.